

Weitai Kang

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Research Summary

I build large multimodal models and the AI agents that harness them — spanning video understanding, visual grounding (2D / GUI / 3D), computer-use agents, and image editing. 9 first-author / co-first-author accepted papers at CVPR, ICCV, ICLR, ECCV, NeurIPS, ACM MM and ACCV.

Education

University of Illinois Chicago

Ph.D. Student, Computer Science

Chicago, IL

Jan. 2025 – May 2027 (expected)

- Research Assistant, advised by Prof. [Yan Yan](#); visiting scholar with Prof. [Mubarak Shah](#) (UCF).

Illinois Institute of Technology

Ph.D. Student, Computer Science (transferred to UIC with advisor)

Chicago, IL

Aug. 2022 – Dec. 2024

- Research Assistant, advised by Prof. [Yan Yan](#).

Sun Yat-sen University

B.S., Information and Computing Science (School of Mathematics)

Guangzhou, China

Sep. 2017 – July 2022

- GPA: 3.7/4.0. Outstanding Student Scholarship, all four years.

Employment Experience

Netflix — Research Intern

Video Understanding

May 2026 – Aug. 2026

Los Gatos, CA

- Published **HeroFrame-Bench**, the first benchmark for movie hero-frame selection, using Rubric-Ranking Co-Evolution to jointly refine movie-specific criteria and reference rankings; scaled to 204 movies, 2,031 chains, and 1,970 rubrics, raising VLM-human agreement to **83.78%**.
- Published **CueKFS**, an agentic keyframe selection method for long-video understanding that iteratively retrieves evidence and revises visual cues; achieved SOTA in all 27 evaluated settings with prior results (up to **+4.54%**) using a median of two reasoning-VLM calls.

Adobe — Research Intern

Visual Grounding and Image Editing

May 2025 – May 2026

San Jose, CA / Chicago, IL

- Published **VGent** (CVPR 2026), a modular architecture decoupling reasoning from box prediction — new SOTA on multi-target visual grounding (**+20.6% F1**) at constant inference latency.
- Published **Inline Critic** (arXiv 2026), a learnable critic token that steers a frozen image editor mid-forward-pass — **81.9 on KRIS-Bench**, the strongest open model, surpassing GPT-4o.

Sony AI — Research Intern

2D Large Multimodal Models

Oct. 2024 – Dec. 2024

Chicago, IL

- Published **ExpVG** (arXiv 2025), a design-space study of visual grounding in MLLMs — **+7.0%** on RefCOCOg.

Tencent — Machine Learning Engineer Intern

Human Pose Detection

Oct. 2021 – July 2022

Shenzhen, China

- Developed human-pose detection models for large-scale image pipelines.

SenseTime — Research Intern

Video Super-Resolution

July 2021 – Sep. 2021

Shenzhen, China

- Researched transformer-based methods for video super-resolution.

Selected Publications

First author unless noted; * equal contribution. Papers and code on [website](#).

- [1] **Weitai Kang**, Hanieh Deilamsalehy, Yumo Xu, Dewang Sultania, Serdar Cellat, Yan Yan, “HeroFrame-Bench: Reference-Anchored Evaluation via Rubric-Ranking Co-Evolution for Movie Hero Frame Selection”, under review.
A frame selection benchmark built with Rubric-Ranking Co-Evolution; 204 movies, 2,031 reference chains, 1,970 learned rubrics, and 83.78% VLM-human agreement.
- [2] **Weitai Kang**, Hanieh Deilamsalehy, Yumo Xu, Dewang Sultania, Serdar Cellat, Yan Yan, “CueKFS: Agentic Cue-Driven Keyframe Selection for Long Video Understanding”, under review.
An agentic keyframe selection method that iteratively retrieves evidence from long video and revises visual cues; SOTA in all 27 evaluated settings with prior results, with gains up to +4.54%.
- [3] **Weitai Kang**, Jason Kuen, Mengwei Ren, Zijun Wei, Yan Yan, Kangning Liu, “VGent: Visual Grounding via Modular Design for Disentangling Reasoning and Prediction”, **CVPR 2026**.
Decouples high-level reasoning from low-level box prediction; +8.2% gIoU / +5.8% cIoU on visual-reference grounding.
- [4] **Weitai Kang**, Xiaohang Zhan, Yizhou Wang, Mang Tik Chiu, Jason Kuen, Kangning Liu, Yan Yan, “Inline Critic Steers Image Editing”, **NeurIPS 2026**
A learnable critic token that refines edits inside a frozen editor’s forward pass.
- [5] **Weitai Kang**, Bin Lei, Gaowen Liu, Caiwen Ding, Yan Yan, “GuirIVG: Incentivize GUI Visual Grounding via Empirical Exploration on Reinforcement Learning”, **ICLR 2026**.
Rule-based RL matches SFT trained on 10M+ samples using only 5.2K; +17.2% on ScreenSpotPro.
- [6] **Weitai Kang**, Haifeng Huang, Yuzhang Shang, Mubarak Shah, Yan Yan, “Robin3D: Improving 3D Large Language Model via Robust Instruction Tuning”, **ICCV 2025**.
A 1M-sample synthetic instruction engine; SOTA across 5 3D benchmarks with no task-specific fine-tuning.
- [7] **Weitai Kang**, Mengxue Qu, Jyoti Kini, Yunchao Wei, Mubarak Shah, Yan Yan, “Intent3D: 3D Object Detection in RGB-D Scans based on Human Intention Language”, **ICLR 2025**.
Introduced the 3D Intention Grounding task and a 45K-intention benchmark; up to +11.2% Top-1 over prior SOTA.
- [8] Bin Lei*, **Weitai Kang***, Zijian Zhang, Winson Chen, Xi Xie, Shan Zuo, Mimi Xie, Ali Payani, Mingyi Hong, Yan Yan, Caiwen Ding, “InfantAgent-Next: A Multimodal Generalist Agent for Automated Computer Interaction”, **NeurIPS 2025**.
Modular multimodal generalist agent; higher OSWorld accuracy than Claude Computer Use.
- [9] **Weitai Kang**, Gaowen Liu, Mubarak Shah, Yan Yan, “SegVG: Transferring Object Bounding Box to Segmentation for Visual Grounding”, **ECCV 2024**.
Recast box labels as pixel-level segmentation supervision; SOTA across 5 visual-grounding datasets.
- [10] **Weitai Kang**, Luowei Zhou, Junyi Wu, Changchang Sun, Yan Yan, “Visual Grounding with Attention-Driven Constraint Balancing (AttBalance)”, **ACM MM 2025**.
Attention-regulation framework; consistent gains across 5 grounding models and 4 benchmarks (+7.1% on RefCOCOg).
- [11] **Weitai Kang**, Weiming Zhuang, Zhizhong Li, Yan Yan, Lingjuan Lyu, “ExpVG: Investigating the Design Space of Visual Grounding in Multimodal Large Language Model”, **ACCV 2026**.
Empirical study on Multimodal Large Language Model design for Visual Grounding.
- [12] **Weitai Kang**, Mengxue Qu, Yunchao Wei, Yan Yan, “ACTRESS: Active Retraining for Semi-supervised Visual Grounding”.
A semi-supervised method for Visual Grounding.

Additional Publications

- [13] Yuzhang Shang, Bingxin Xu, **Weitai Kang**, Mu Cai, Yuheng Li, Zehao Wen, Zhen Dong, Kurt Keutzer, Yong Jae Lee, Yan Yan, “INTP: Interpolating Video-LLMs Toward Longer-sequence LMMs in a Training-free Manner”.
- [14] Junyi Wu, **Weitai Kang**, Hao Tang, Yuan Hong, Yan Yan, “On the Faithfulness of Vision Transformer Explanations”, **CVPR 2024**.
- [15] Junyi Wu, Bin Duan, **Weitai Kang**, Hao Tang, Yan Yan, “Token Transformation Matters: Towards Faithful Post-hoc Explanation for Vision Transformer”, **CVPR 2024**.
- [16] Wenxin Chen, Mengxue Qu, **Weitai Kang**, Yan Yan, Yao Zhao, Yunchao Wei, “3DResT: A Strong Baseline for Semi-Supervised 3D Referring Expression Segmentation”, **IEEE TMM 2025**.

[17] Bin Lei, Yuchen Li, Yiming Zeng, Tao Ren, Yi Luo, Tianyu Shi, Zitian Gao, Zeyu Hu, **Weitai Kang**, Qiuwu Chen, “Infant Agent: A Tool-Integrated, Logic-Driven Agent with Cost-Effective API Usage”.

Skills

Research: Multimodal LLMs/VLMs · Long-Video Understanding · Visual Grounding (2D/GUI/3D) · Computer-Use & Generalist Agents · Image Editing · Reinforcement Learning (RLVR, rule-based RFT) · Data Engines & Synthetic Data · Benchmark Design

Systems: PyTorch · HuggingFace Transformers · distributed / multi-GPU training · Python, C/C++ · Linux, Git

Languages: English (proficient) · Mandarin (native) · Cantonese (native)